1 **(a)** 17.5 (or $17\frac{7}{15}$ or 17.46...) **[B1]**

(b) 17 **[B1]**

(c) Upper quartile = 21 and Lower quartile = 15 **[M1]**

6 **[A1]**

[Total Marks: 4]

2 **(a)** **(i)** 3.0476... **[M1]**

3.05 **[A1]**

(ii) 3 **[B1]**

(iii) Upper quartile = 4 and Lower quartile = 2 identified **[M1]**

2 **[A1]**

[Total Marks: 5]

3 **(a)** 16 **[B2]**

If B0, M1 for $\sum x = 240$ seen

(b) 15 **[B1]**

(c) 6 **[B2]**

If B0, M1 for $Q_1 = 13$ and $Q_3 = 19$ seen

[Total Marks: 5]

4 **(a)** 5 **[B1]**

(b) 5 **[B1]**

(c) 5 **[B1]**

(d) 4 **[B1]**

(e) 6 **[B1]**

[Total Marks: 5]

5 **(a)** **(i)** 27 **[B1]**

(ii) 26 **[B1]**

(iii) Lower quartile = 20 **[B1]**

Upper quartile = 33 **[B1]**

(b) 13 **[B1]**

(c) $(16 \times 27.5) - (15 \times 27) = 440 - 405$ **[M1]**

35 **[A1]**

[Total Marks: 7]

6 **(a)** $150 < h \leq 160$ **[B1]**

(b) Uses midpoints $135, 145, 155, 165, 175$ **[M1]**

152 **[A1]**

[Total Marks: 3]

7 **(a)** $20 < t \leq 30$ **[B1]**

(b) Use of correct midpoints: $5, 15, 25, 35, 50$ **[M1]**

$\Sigma fx = 14(5) + 28(15) + 35(25) + 18(35) + 5(50)$ or 2245 seen **[M1]**

22.45 **[A1]**

(c) The exact times are not known so the midpoints of the intervals are used. **[B1]**

[Total Marks: 5]

8 **(a)** Use of correct mid-interval values $(5, 15, 25, 35, 50)$ **[M1]**

$\Sigma ft = 5(12) + 15(25) + 25(28) + 35(11) + 50(4)$ or 1720 **[M1]**

21.5 **[A1]**

(b) The exact times are not known / the data is grouped so mid-interval values are used. **[B1]**

(c) The mean will decrease because 12 is less than the current mean (21.5). **[B1]**

[Total Marks: 5]

9 **(a)** 21.25 **[B2]**

If B0, M1 for $\sum fx = 1700$ or midpoints $5, 15, 25, 35, 45$ seen

(b) The exact times are not known / the midpoints of each class are used as an estimate. **[B1]**

(c) $120 \times 24.5 = 2940$ **[M1]**

$(1700 + 2940) \div 200$ **[M1]**

23.2 **[A1]**

[Total Marks: 6]

10 **(a)** **(i)** 20.75 **[B1]**

(ii) 21 **[B1]**

(iii) Upper quartile = 22 and lower quartile = 20 **[M1]**

2 **[A1]**

(b) **(i)** Mid-interval values: $1, 3.5, 7.5, 15$ **[B1]**

$\dfrac{8(1)+17(3.5)+15(7.5)+10(15)}{50} = \dfrac{330}{50}$ **[M1]**

6.6 **[A1]**

(ii) $2 < m \leq 5$ **[B1]**

[Total Marks: 8]

11 **(a)** $3+x+y+6+4+1 = 35$ leading to $x+y = 21$ **[M1]**

Expression for total goals: $0(3)+1(x)+2(y)+3(6)+4(4)+5(1)$ or $x+2y+39$ **[M1]**

$\dfrac{x+2y+39}{35} = 2.2$ leading to $x+2y = 38$ **[M1]**

$x = 4$ and $y = 17$ **[A1]**

(b) 2 **[B1]**

[Total Marks: 5]

12 **(a)** $\dfrac{0\times 6+1\times x+2\times 8+3\times 4+4\times 2}{6+x+8+4+2} = 1.5$ **[M1]**

$\dfrac{x+36}{x+20} = 1.5$ **[M1]**

$x+36 = 1.5x+30 \implies 0.5x = 6 \implies x = 12$ **[A1]**

(b) 1 **[B1]**

(c) 1 **[B2]**

If B0, M1 for $Q_1 = 1$ and $Q_3 = 2$ seen

[Total Marks: 6]

13 **(a)** Total frequency $= 10+k+2k+3+7+5 = 3k+25$ **[M1]**

Total score $= 10(1)+k(2)+(2k+3)(3)+7(4)+5(5) = 8k+72$ **[M1]**

$\dfrac{8k+72}{3k+25} = 2.8$ **[M1]**

$8k+72 = 8.4k+70 \implies 0.4k = 2 \implies k = 5$ **[A1]**

(b) **(i)** 3 **[B1]**

(ii) 1.5 **[B1]**

(iii) 4 **[B1]**

[Total Marks: 7]

14 **(a)** $8+x+y+6+2 = 40 \implies x+y = 24$ **[B1]**

$\dfrac{0\times 8+1\times x+2\times y+3\times 6+4\times 2}{40} = 1.55$ **[M1]**

$x + 2y + 26 = 62 \implies x + 2y = 36$ **[A1]**

(b) Subtracts equations: $(x + 2y) - (x + y) = 36 - 24 \implies y = 12$ **[M1]**

$x = 12, y = 12$ **[A1]**

(c) **(i)** 1.5 **[B1]**

(ii) 1 **[B1]**

[Total Marks: 7]

15 **(a)** $34 + p + 26 + q + 8 = 100$ simplifying to $p + q = 32$ **[B1]**

(b) **(i)** $\sum fm = 34(2.5) + 7.5p + 26(15) + 25q + 8(40) = 7.5p + 25q + 795$ **[M1]**

$\frac{7.5p + 25q + 795}{100} = 12.45$ giving $7.5p + 25q = 450$ **[M1]**

Multiplying by $\frac{2}{5}$ gives $3p + 10q = 180$ **[A1]**

(ii) $p = 20$ **[B1]**

$q = 12$ **[B1]**

(c) **(i)** 24.25 **[B1]**

(ii) 24 **[B1]**

(iii) Upper quartile = 27 and lower quartile = 22 **[M1]**

5 **[A1]**

(d) Revised $\sum fm = 1245 - 12(25) + 291 = 1236$ **[M1]**

$\frac{1236}{100} = 12.36$ **[A1]**

[Total Marks: 12]